

Diagnosis and management of type 2 diabetes

Part of the HEARTS technical package for cardiovascular disease management in primary health care

Diagnosis and management of type 2 diabetes is based on WHO guidance on diagnosis, classification and management of diabetes. It is aligned with the WHO Package of Essential Noncommunicable Disease Interventions in Primary Health Care (WHO-PEN). It can be used independently or in conjunction with the HEARTS technical package developed to improve cardiovascular health.

Target users may vary, based on context, existing health systems and national priorities.

The people who will find the modules most useful are:

? National level ? Ministry of Health NCD policymakers responsible for:

- developing strategies, policies and plans related to service delivery of diabetes
- setting national targets on diabetes, monitoring progress and reporting

? Subnational level ? Health/NCD programme managers responsible for:

- planning, training, implementing and monitoring service delivery

? Primary care level ? Facility managers and primary health care staff and trainers responsible for:

- assigning tasks, organising training and ensuring the facility is running smoothly
- clinical management of people with type 2 diabetes
- collecting facility-level data on indicators of progress towards diabetes targets.

Introduction

Definition of diabetes

Diabetes mellitus, commonly known as diabetes, is a group of metabolic disorders characterized by the presence of hyperglycaemia in the absence of treatment. The heterogeneous aetiopathology includes defects in insulin secretion, insulin action, or both. The long-term specific complications of diabetes include retinopathy, nephropathy, and neuropathy. People with diabetes are also at increased risk of other diseases, including cardiac, peripheral arterial and cerebrovascular disease, cataracts, erectile dysfunction, and nonalcoholic fatty liver disease. They are also at an increased risk of some infectious diseases such as tuberculosis, and are likely to experience poorer outcomes.

Epidemiology and global burden of diabetes

Diabetes is found in every population in the world and in all regions, including rural parts of low- and middle-income countries. WHO estimates there were 422 million adults with diabetes worldwide in 2014. The age-adjusted prevalence in adults rose from 4.7% in 1980 to 8.5% in 2014, with the greatest increase in low- and middle-income countries.

Aetio-pathology of diabetes

The underlying characteristic common to all forms of diabetes is the dysfunction or destruction of pancreatic beta-cells. These cells are not replaced, as the human pancreas seems incapable of renewing beta-cells after the age of 30 years. Many mechanisms can lead to a decline in function or the complete destruction of beta-cells. These mechanisms include genetic predisposition and abnormalities, epigenetic processes, insulin resistance, auto-immunity, concurrent illnesses, inflammation, and environmental factors.

Most common types of diabetes and their risk factors

Risk factors for type 2 diabetes:

- ? overweight/obesity
- ? physical inactivity
- ? age
- ? diabetes in first degree relatives
- ? history of gestational diabetes
- ? cardiovascular disease and its risk factors
- ? ethnicity (South Asian, Afro-Caribbean, Hispanic)

The most common type of diabetes mellitus is type 2 diabetes (T2DM). The majority of people with T2DM are overweight or obese, which either causes or aggravates insulin resistance.

Many of those who are not obese by BMI criteria have a higher proportion of body fat distributed predominantly in the abdominal region, indicating visceral adiposity, compared to people without diabetes.

However, in some populations, such as Asians, beta-cell dysfunction appears to be a more prominent feature than in populations of European descent. This is also observed in thinner people from low- and middle-income countries such as India, and among people of Indian descent living in high-income countries.

Risk factors for type 1 diabetes:

? certain genetic haplotypes

? unknown environmental factors

Type 1 diabetes is much less common, the risk being highest in populations of European origin. This module is specifically for management of type 2 diabetes, as type 1 is usually not managed in primary care facilities.

Classification of diabetes

The WHO classification of diabetes is presented in Table 1. It prioritizes clinical care and guides health professionals in choosing appropriate treatments at the time of diabetes diagnosis, providing practical guidance to clinicians in assigning a type of diabetes to individuals at the time of diagnosis.

Table 1 Classification of diabetes

Type 1 diabetes: Beta-cell destruction (mostly immune-mediated) and absolute insulin deficiency; onset most common in childhood and early adulthood.

Type 2 diabetes: Most common type, various degrees of beta-cell dysfunction and insulin resistance; commonly associated with overweight and obesity.

Hybrid forms of diabetes:

- Slowly evolving, immune-mediated diabetes of adults: Similar to slowly evolving type 1 in adults but more often has features of the metabolic syndrome, a single GAD autoantibody, and retains greater beta-cell function.
- Ketosis-prone type 2 diabetes: Presents with ketosis and insulin deficiency but later does not require insulin; common episodes of ketosis, not immune-mediated.

Other specific types:

- Monogenic defects of beta-cell function or insulin action.
- Diseases of the exocrine pancreas (trauma, tumour, inflammation).
- Endocrine disorders (excess secretion of antagonist hormones).
- Drug- or chemical-induced impairment.
- Infection-related diabetes (viral destruction).
- Uncommon specific forms of immune-mediated diabetes.
- Genetic syndromes associated with diabetes.
- Unclassified diabetes.

Hyperglycaemia first detected during pregnancy:

- Diabetes mellitus in pregnancy (Type 1 or 2 diagnosed during pregnancy).
- Gestational diabetes mellitus (Hyperglycaemia below diagnostic thresholds for diabetes in pregnancy).

1 Clinical manifestations of diabetes and diagnostic criteria

Clinical manifestations

Diabetes may present with characteristic symptoms and signs (Table 2). It is estimated that a significant percentage of cases of type 2 diabetes (T2DM) (30% to 80%, depending on the country) are undiagnosed. The most severe clinical manifestations are ketoacidosis or a non-ketotic hyperosmolar state that may lead to dehydration, coma and, in the absence of effective treatment, death. However, symptoms in T2DM are often not severe, or may be absent, owing to the slow pace at which the hyperglycaemia is worsening. As a result, in the absence of biochemical testing, hyperglycaemia sufficient to cause pathological and functional changes may be present for a long time before a diagnosis is made, resulting in the presence of complications at diagnosis.

Type 1 diabetes is more likely to present with symptoms and its onset is typically in children and young adults. However, the type of diabetes cannot always be determined at diagnosis, and initial treatment decisions should be based on clinical presentation and plasma glucose values.

Table 2 Symptoms and signs of diabetes

Symptoms of diabetes:

- ? thirst
- ? frequent urination
- ? blurring of vision
- ? fatigue

Signs of diabetes:

- ? unintentional weight loss
- ? signs of acute metabolic deterioration (dehydration, Kussmaul's respiration, vomiting, altered consciousness)
- ? clinical signs of chronic complications (coronary disease, stroke, kidney disease, vision loss, diabetic foot)

Diagnostic criteria for diabetes

Diagnosis of diabetes is based on values of plasma glucose or glycated haemoglobin (HbA1c). Diagnostic cut-off values are presented in Table 3.

Table 3 Diagnostic criteria for diabetes:

- ? Fasting venous or capillary plasma glucose: ≥ 7.0 mmol/L (126 mg/dL) [Least costly, requires fasting]
- ? 2-hour post-load venous plasma glucose: ≥ 11.1 mmol/L (200 mg/dL)
- ? 2-hour post-load capillary plasma glucose: ≥ 12.2 mmol/L (220 mg/dL)
- ? Random plasma glucose: ≥ 11.1 mmol/L (200 mg/dL) [To be used only in presence of symptoms]
- ? HbA1c: 6.5% (48 mmol/mol) [Less intra-individual variability, does not require fasting]

Diagnostic tests:

Venous plasma glucose is the standard method for measuring and reporting. Fasting values for venous and capillary plasma glucose are identical.

? In asymptomatic people, repeat the test to confirm the diagnosis on a subsequent day.

? If plasma glucose ≥ 18 mmol/L (325 mg/dL), or symptoms are present, measure urine ketones.

? If plasma glucose measurement is not possible, urine glucose testing can be used to confirm suspicion in symptomatic people.

2 Management of diabetes

Type 2 diabetes is a progressive illness, with insulin secretion decreasing over time. Introduction of oral hypoglycaemic agents (OHA) will often be necessary in patients treated with diet and physical activity only, and further intensification with insulin might be needed as the illness progresses.

Blood glucose management protocol:

The blood glucose management protocol is recommended for patients with established or newly diagnosed type 2 diabetes.

Non-pharmacological management:

A healthy diet to achieve or maintain normal body weight and regular physical activity are the mainstay of diabetes management.

? Eat a healthy balanced diet applicable to general population.

? Overweight patients should reduce weight by reducing calorie intake.

? Practise regular daily physical activity appropriate for physical capabilities (at least 150 minutes of moderate or vigorous aerobic activity per week, spread over at least 3 days).

? Avoidance of tobacco use and harmful use of alcohol.

Pharmacological management:

Control of blood glucose levels (glycaemia):

Initial treatment:

? Metformin does not cause weight gain or hypoglycaemia and is the recommended initial treatment for people who do not achieve desired glycaemic control with diet and physical activity.

? A second-generation sulfonylurea (preferably gliclazide) can be used as initial treatment when metformin is contraindicated or not tolerated.

? Other pharmacological agents have not been shown to be superior to metformin or sulfonylurea for initial treatment.

Metformin

Metformin is contraindicated in:

- ? chronic kidney disease (eGFR < 30 mL/minute/1.73m²)
- ? severe reduced liver function
- ? acute cardiac insufficiency
- ? respiratory insufficiency
- ? alcohol abuse
- ? history of lactic acidosis

Intensification of treatment when metformin alone fails to control glycaemia:

- ? Add a second-generation sulfonylurea (preferably gliclazide) to metformin.
- ? In hyperglycaemic patients with symptoms, give a sulfonylurea or refer for insulin treatment.

Glibenclamide is not recommended in:

- ? people aged 60 years or older
- ? severe liver disease
- ? patients for whom hypoglycaemia is a concern (risk of falls, impaired awareness, living alone)
- ? people who drive or operate machinery

Intensification when metformin and sulfonylurea fail:

- ? Refer for insulin treatment or add human insulin to oral medication.
- ? If insulin is unsuitable, a DPP-4 inhibitor, SGLT-2 inhibitor or TZD may be added.

Fig. 1 Protocol for control of blood glucose in type 2 diabetes

TEST ADULTS who have symptoms of diabetes with fasting or random plasma glucose (FPG or RPG).

TEST ASYMPTOMATIC ADULTS who are 40+ years old and BMI ≥ 25 (FPG).

? If FPG ≥ 7 mmol/L and < 18 mmol/L or RPG ≥ 11.1 mmol/L:

- If asymptomatic, repeat test on subsequent day.
- Counsel on diet and physical activity.
- REVIEW IN 1 MONTH.
- If goal not achieved: BEGIN METFORMIN 500 mg daily.
- REVIEW IN 3 MONTHS -> INCREASE METFORMIN to 1000 mg 1x daily.
- REVIEW IN 3 MONTHS -> INCREASE METFORMIN to 1000 mg 2x daily.
- REVIEW IN 3 MONTHS -> ADD gliclazide 80 mg 1x daily.
- REVIEW IN 3 MONTHS -> ADD gliclazide 80 mg 2x daily.
- REVIEW IN 3 MONTHS -> REFER to higher level or BEGIN INSULIN.

? If FPG ≥ 15 mmol/L and SYMPTOMATIC or FPG/RPG > 18 mmol/L:

- TEST urine ketones.
- If ketones $\geq 2+$: REFER to higher level of care.
- If ketones $< 2+$: GIVE METFORMIN 1000 mg 2x daily and GLICLAZIDE 80 mg 2x daily.
- REVIEW in 3-5 days.

Control of blood lipids and pressure:

Blood pressure should be measured at every visit. Hypertension treatment is indicated in people with diabetes when systolic blood pressure ≥ 130 mmHg or diastolic blood pressure ≥ 80 mmHg. Statins are recommended for all people with type 2 diabetes 40 years old or older.

Monitoring of glycaemic control:

Glycated haemoglobin (HbA1c) is commonly used in clinical practice to monitor glycaemic control (measures average plasma glucose over preceding 8 to 12 weeks).

If HbA1c is not available:

? Fasting plasma glucose (FPG) values can be used.

? A combination of FPG and postprandial plasma glucose (2 hours after breakfast) is more informative.

Targets for glycaemic control:

? The majority of patients can be expected to aim for an HbA1c of 7.0% (53 mmol/mol).

? The HbA1c target can be relaxed (e.g. to $< 8\%$ or < 64 mmol/mol) in people with frequent severe hypoglycaemia, advanced complications or low life-expectancy.

? If HbA1c is not available, FPG ≤ 7.0 mmol/L (126 mg/dL) and postprandial PG ≤ 9.0 mmol/L (160 mg/dL) serve as surrogates.

3 Prevention and management of complications of diabetes

Acute complications of diabetes:

Two important acute complications are hypoglycaemia and hyperglycaemic emergencies.

Hypoglycaemia:

Hypoglycaemia (abnormally low blood glucose) is a frequent iatrogenic complication in diabetic patients, occurring particularly in patients receiving sulfonylurea or insulin. It can cause loss of consciousness and coma and is potentially life-threatening.

Defined at plasma glucose of ≤ 3.9 mmol/L (70 mg/dL).

Table 4 Symptoms and signs of hypoglycaemia

Symptoms:

? headache, hunger, irritability, anxiety, paraesthesias, palpitations

Signs:

? sweating, trembling, difficulty in speaking, confusion, ataxia, stupor, pallor, seizures, coma

Management of hypoglycaemia:

- ? If conscious and able to ingest food or drink: ingest 15-20 g of glucose (or sugar-sweetened soft drink, 1-2 teaspoons sugar, 5-6 hard candies). Recheck in 15 minutes. Follow by a small meal with complex carbohydrate.
- ? Unconscious patients or plasma glucose ≤ 2.8 mmol/L (50 mg/dL): give hypertonic glucose (dextrose) intravenously (20-50 mL of 50% glucose over 1-3 minutes).

Hyperglycaemic emergencies:

Diabetic ketoacidosis (DKA) and hyperosmolar hyperglycaemic state (HHS) are life-threatening conditions.

Symptoms/signs: nausea, vomiting, abdominal pain, Kussmaul's breathing, altered consciousness.

Table 5 Biochemical characteristics of DKA and HHS:

? DKA: Plasma glucose ≥ 13.9 mmol/L (250 mg/dL), Urine ketones Positive.

? HHS: Plasma glucose ≥ 33.3 mmol/L (600 mg/dL), Urine ketones Negative (or weakly positive).

Management of hyperglycaemic emergencies:

? Refer to hospital all patients with suspected DKA or HHS.

? Infuse isotonic saline (0.9% NaCl) at a rate of 1000 mL in first 2 hours, continue 1000 mL every 4 hours.

Screening and management of chronic complications of diabetes

Microvascular complications:

Longstanding diabetes with uncontrolled blood glucose levels can lead to diabetic retinopathy, nephropathy, neuropathy, and diabetic foot complications.

Diabetic eye disease:

? Diabetic retinopathy is a highly specific microvascular complication of diabetes and leading cause of blindness.

? Risk factors: duration of diabetes, poor glycaemic control, hypertension, diabetic kidney disease, dyslipidaemia.

? Diagnosis: fundus examination after pupil dilation (microaneurysms, haemorrhages, hard exudates, cotton-wool spots).

? Screening: screen upon diagnosis, and biennially thereafter.

Diabetic kidney disease:

Diabetic kidney disease is a microvascular complication characterized by decline in eGFR, raised blood pressure, and high risk of CVD.

? Defined by albuminuria and/or decreased eGFR ($< 60 \text{ mL/min per } 1.73 \text{ m}^2$ on at least two occasions 1-3 months apart).

? Risk factors: poor glycaemic control, elevated blood pressure, genetic susceptibility.

Table 6 Categories of albuminuria (measured by ACR):

? Normal to mildly increased: $< 30 \text{ mg/g}$ ($< 3 \text{ mg/mmol}$)

? Moderately increased: $30\text{--}300 \text{ mg/g}$ ($3\text{--}30 \text{ mg/mmol}$)

? Severely increased: $> 300 \text{ mg/g}$ ($> 30 \text{ mg/mmol}$)

Screening & management:

? Screen once a year with spot urine ACR or eGFR.

? Maintain blood pressure levels at $< 130/80 \text{ mmHg}$ with an ACE inhibitor and add thiazide diuretic if required.

Diabetic neuropathy:

Nerve damage in diabetes includes distal symmetrical peripheral neuropathy (sensory) and autonomic neuropathy.

Risk factors: duration of diabetes, poor glycaemic control, age.

Diagnosis:

? Peripheral: sensory loss, unsteadiness, burning/tingling/numbness.

? Autonomic: lack of awareness of hypoglycaemia, orthostatic hypotension, resting tachycardia, erectile dysfunction.

Management:

? Exclude non-diabetic causes (B12 deficiency, alcohol, hypothyroidism).

? Refer painful peripheral neuropathy or autonomic neuropathy to specialized care.

? Improve glycaemic control.

Foot problems in diabetes:

Diabetic foot ulcers result from peripheral neuropathy (loss of protective sensation) and peripheral artery disease (PAD).

Macrovascular complications:

Coronary heart disease, cerebrovascular disease, and peripheral vascular disease.

Risk factors: age, family history, obesity, chronic kidney disease, hypertension, dyslipidaemia, smoking.

Blood pressure control:

Target blood pressure < 130/80 mmHg using Thiazide diuretics and ACE inhibitors.

Control of blood lipids:

Statins are recommended for all patients of 40 years of age and older with diabetes.

Antiplatelet treatment:

75-100 mg of acetylsalicylic acid (aspirin) daily for secondary prevention of CVD events in patients who survived a CVD event.

4 Criteria for referral to higher levels of care

Urgent (same day) referral should occur if:

- ? urine ketones > 2+, or no improvement in glycaemia ≥ 18 mmol/L after management with metformin/gliclazide
- ? suspicion of DKA or HHS
- ? hypoglycaemia unresolved by treatment
- ? clinical suspicion of type 1 diabetes in newly diagnosed patient
- ? symptoms/signs of coronary heart disease and stroke
- ? recent deterioration of vision
- ? blood pressure > 200/> 110 mmHg
- ? blood pressure > 180/> 110 mmHg with headache, shortness of breath, blurred vision
- ? infected foot ulcer, gangrene, critical limb ischaemia
- ? anuria or eGFR < 30 mL/min/1.73 m²

Non-emergency referral:

- ? glycaemia goal not achieved despite compliance
- ? eGFR 30-59 mL/min/1.73 m²
- ? moderately and severely increased albuminuria
- ? BP > 130/80 mmHg despite two medications
- ? total cholesterol > 8 mmol/L (310 mg/dL).

5 Monitoring of processes and outcomes

Biochemical parameters:

? HbA1c $\leq$ 7.0% (53 mmol/mol)

? Surrogate FPG $\leq$ 7.0 mmol/L (126 mg/dL) and postprandial PG $\leq$ 9.0 mmol/L (160 mg/dL)

Table 7 Assessment of effectiveness of diabetes services:

? Number of patients being treated for diabetes (Reporting: monthly)

? Control rate among people treated for diabetes (HbA1c $<$ 7.0% or FPG $<$ 7.0 mmol/L) (Reporting: every 6 months)

? Complications due to diabetes (foot, nephropathy, retinopathy, neuropathy, CVD) (Reporting: annually)

Annex 1: Protocol for treatment of type 2 diabetes mellitus with insulin

- ? START with 10 units intermediate-acting insulin (NPH) at bedtime.
- ? Continue metformin and sulfonylurea.
- ? Counsel patient on hypoglycaemia and hyperglycaemia symptoms.
- ? REVIEW in 3 days.
- ? If FBG > 7 mmol/L: INCREASE INSULIN dosage by 1?2 units at 3-day intervals until FBG = 4?7 mmol/L.
- ? Do not increase insulin if nocturnal hypoglycaemia occurs.
- ? If FBG < 4 mmol/L or nocturnal hypoglycaemia occurs: REDUCE INSULIN by 1?2 units.
- ? If FBG 4?7 mmol/L and daytime hypoglycaemia occurs: REDUCE morning gliclazide by 40 mg.
- ? If HbA1c > 7.5%: REFER to higher level of care for intensification.

Annex 2: Prevention, assessment and management of diabetic foot

Diabetic foot problems encompass diabetic foot ulcer and diabetic foot infections (including osteomyelitis). Diabetic foot ulcers result from peripheral neuropathy and peripheral artery disease (PAD).

Assessment of loss of protective sensation (LOPS):

- a) Semmes-Weinstein 10 g monofilament test (3 sites on both feet).
- b) Vibration perception with 128 Hz tuning fork on toe phalanx.
- c) Ipswich Touch Test (light touch to tips of 1st, 3rd, 5th toes).

Table 8 Stratification of risk of diabetic foot:

- ? Low risk: No risk factor except callus alone -> Assess annually.
- ? Moderate risk: Deformity, neuropathy, non-critical ischaemia -> Assess every 3?6 months.
- ? High risk: Previous ulcer/amputation, neuropathy with ischaemia/callus -> Assess every 1?3 months.
- ? Active foot problem: Ulcer, infection, gangrene, Charcot arthropathy -> Urgent referral.